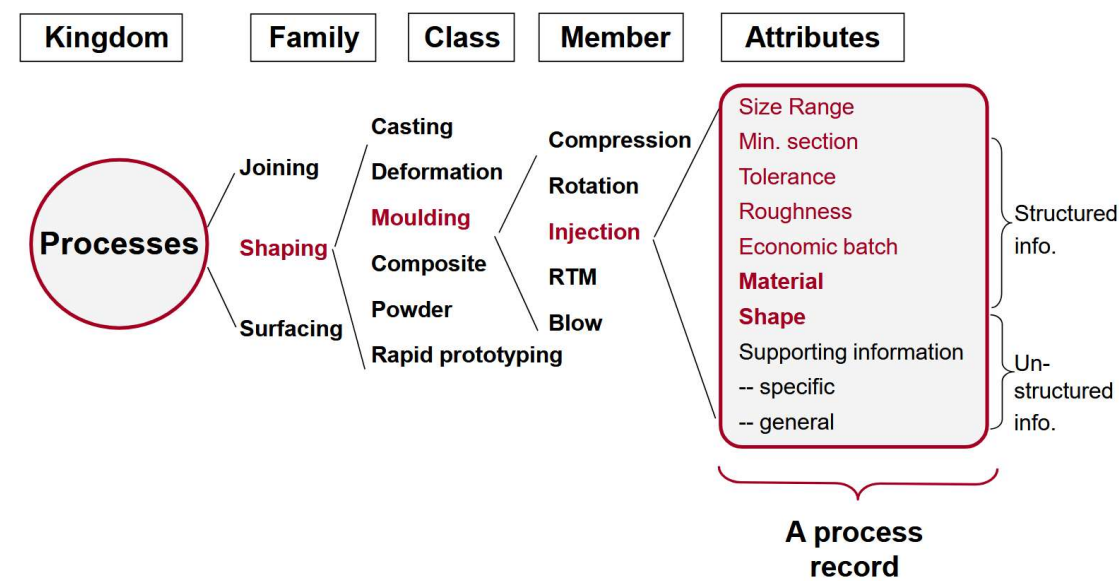
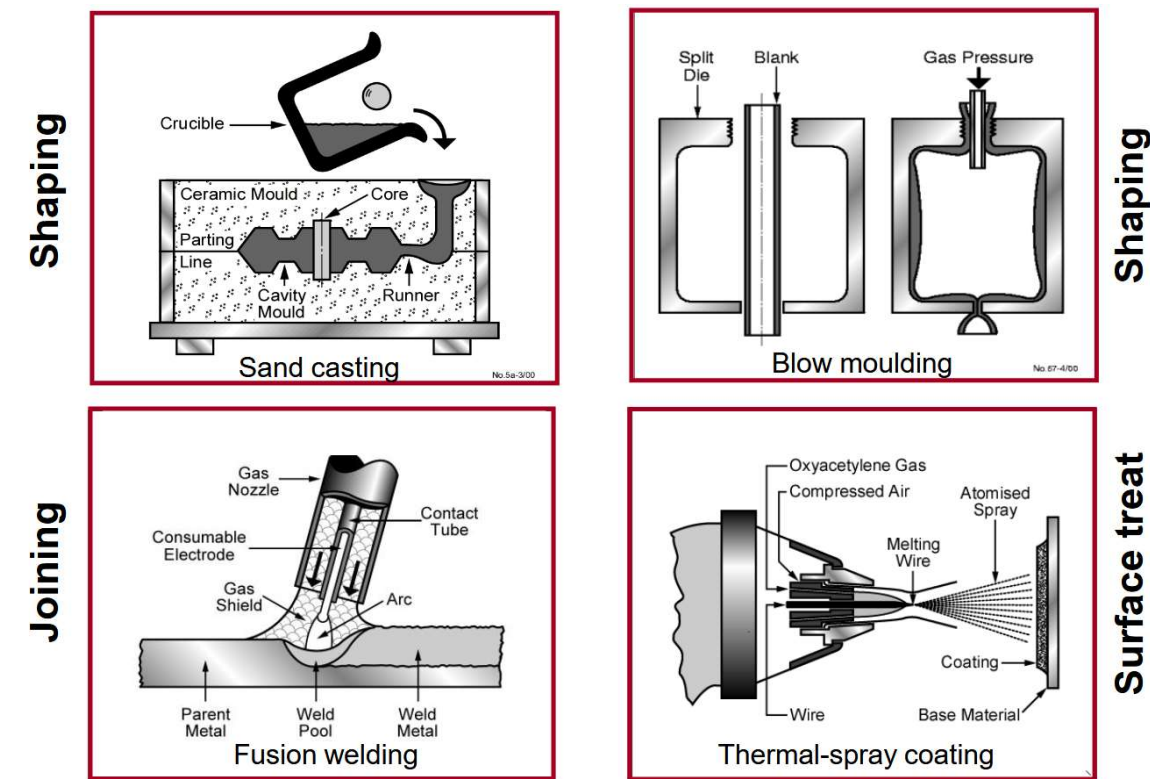
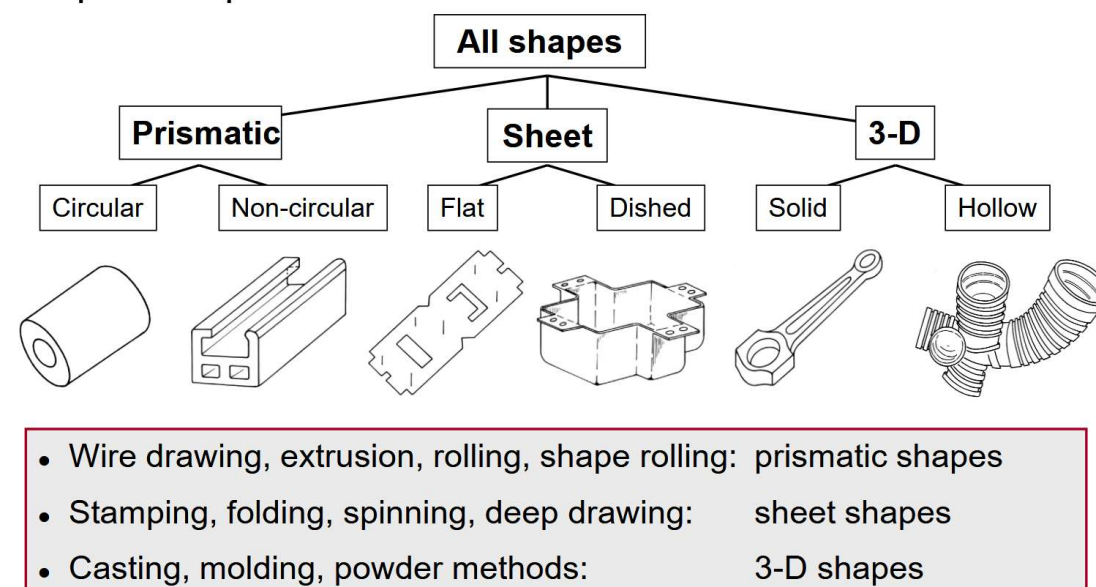


Process : method of shaping, joining, or surface treating a material



Shape classification:

Some processes can make only simple shapes, others, complex shapes.

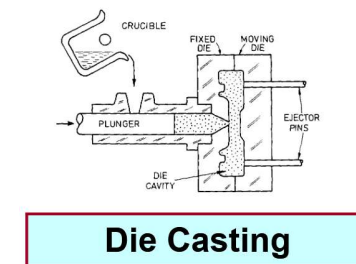
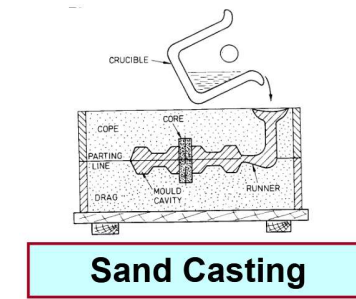


Manufacturing Processes:

Primary:

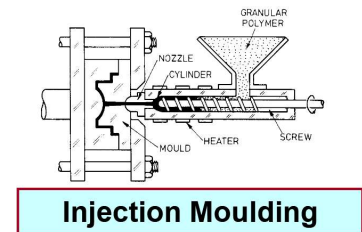
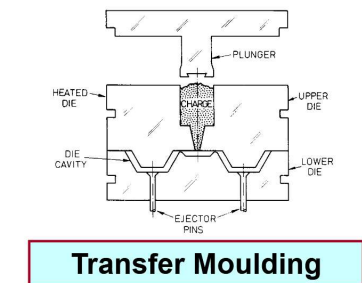
- Casting:

- A liquid is poured into a mould where it solidifies by cooling (Metals) or by reaction (thermosets or elastomers).
- Different from moulding due to the low viscosity of the liquid.
- Gravity casting - mould filled by flow under own weight.
- Centrifugal & pressure-die casting - mould filled under modest pressure.
- Sand moulds - One off ; cheap
- Die casting - Mass production



- Moulding

- Used with higher viscosity fluids - thermoplastics, thermosetting elastomers, glasses
- Hot viscous material pressed or injected into die under considerable pressure.
- Cools & solidifies
- Elaborate shapes can be moulded, but at penalty of complexity in die shape and the way it separates to allow removal.



- Deformation

- Powder

- Special Methods

Secondary:

- Machining

- Heat treatment

- Joining

For metals:

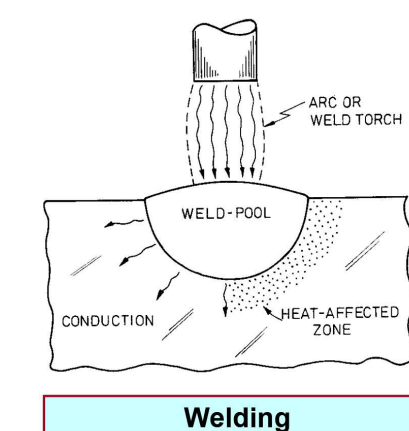
- Bolting
- Riveting
- Welding
- Brazing
- Soldering

For Polymers:

- Snap fasteners
- Thermal bonding
- Ultrasonic welding

Ceramics:

- Can be diffusion bonded to themselves, glasses & metals
- Improved adhesives give new ways of bonding all classes of materials



- Other finishing operations